

PHYS 400

SPECIAL PROBLEMS IN PHYSICS

Guidelines for the Interim Report

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IMPORTANT: Consistency and academic integrity are required for all PHYS 400 tasks. Plagiarism and the use of generative AI will not be tolerated!

All given word counts are suggestions. As long as you find it appropriate, you can write longer or shorter sections, within reason (with the exception of the abstract, which must remain concise).

Introduction

Update your introduction by building on the *Introduction* section from your original proposal. Use this section to show what you have learned more about the subject since your initial submission (**approximately 400-500 words**).

For the interim report, the introduction should:

- Provide one or two paragraphs that clearly state your hypothesis or project question. Briefly explain the specific steps you are taking to test that hypothesis or find the answer to your question.
- Include a proper literature search with **at least 10 citations to peer-reviewed sources**. If you are working on a problem that has been studied before, provide a brief summary of the work others have already done.
- Provide a short summary of your approach, since this report doesn't have a separate *Methods* section. You can base this on a condensed version of the *Methods* section from your original proposal: A single paragraph with a couple of key references is sufficient.

The other important points, based on the project proposal guideline, still remain essential:

- Clearly describe the general research area.
- Discuss the most significant previous studies in your field.
- Explain why this topic is scientifically important.
- Compare different approaches, models, or methods.
- Identify the strengths and limitations of previous work.
- Identify the gap, limitation, or unresolved question (Remember “the observation” discussed in the lectures).
- Clearly connect and organize the text and literature toward your research question.
- Avoid informal language and write the report in an academic style.

This section is NOT a list of summaries. It should be analytical and structured.

Project Progress

In the *Project Progress* section, report on your current progress and outcomes. For each work package, explain what you did, what you found, the challenges you ran into, and how you intend to fix those issues moving forward. (**approximately 800 words**).

For each work package:

- Report on the current status of each work package alongside the findings you have collected during the process.
- Detail any revisions made to your original plan (if any), explain the reasons behind them, and revise your semester plan to reflect these updates.
- Report encountered problems (if any). What are these problems? Have you managed to solve them? If so, how? If you haven't yet, do you have any plans to solve them, or are you going to use an alternative approach? If you have diverged from your original plan, don't forget to report them as revisions to work packages.

Notes about the semester plan:

- Week numbers must be between 1 and 15 (covering the period from the week of February 16 to the week of June 1).
- **Holiday note:** The week of May 25–31 is expected to be a block holiday for *Kurban Bayramı* (Eid al-Adha); this week is not counted in the 15-week schedule.
- The start week must be less than or equal to the end week.
- Avoid overlapping too many tasks unless justified.
- Include time for writing and revision at the end of the semester.
- Extra instructions for creating the timeline in LaTeX are available in the next sections.

Future Outlook

In this section, you are expected to outline your plan (upcoming objectives, milestones, etc.) for the rest of the semester (**approximately 200 words**).

References

- All sources must be directly relevant to your research topic. Wikipedia, blogs, and non-academic websites are **NOT** acceptable as primary sources.
- Every source cited in the text must appear in the reference list.

- The references should be in APA format and sorted alphabetically. In LaTeX, reference formatting and sorting are done automatically.

The following sections remain unchanged from the original project proposal guidelines.

Instructions for LaTeX

How to include a figure

- Place your image file (.png, .jpg, or .pdf) in the same folder as main.tex.
- Use the following structure:

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.6\textwidth]{example.png}
  \caption{Write a descriptive caption.}
  \label{fig:yourlabel}
\end{figure}
```

- Do not manually type figure numbers. Refer to the figure in your text using `Figure~\ref{fig:yourlabel}`.
- The option [h] tries to enforce the location of the picture “here” provided that the space is available. If you want to place the figures at the top use [t] and for bottom use [b].
- The [width=0.6\textwidth] option (following \includegraphics) enables to adjust the image size. If you use 0.6, it means that image size is adjusted for 60% of the horizontal text space, while keeping the aspect ratio constant.
- If an image is taken from a source that requires a citation in the caption, you must use \protect before the \cite command (e.g. \protect\cite{ref1}).

How to include a table

- Use the following structure:

```
\begin{table}[htbp]
  \centering
  \caption{Write a descriptive caption.}
  \label{tab:yourlabel}
  \begin{tabular}{ccc}
    \hline
    Column1 & Column2 & Column3 \\
  \end{tabular}
```

```

\hline
Data1    & Data2    & Data3    \\
Data4    & Data5    & Data6    \\
\hline
\end{tabular}
\end{table}

```

- `\begin{table}[htbp]` allocates the space for the table.
- `\centering` specifies that the table will be centered on the page.
- `\begin{tabular}{ccc}` creates actual table structure. The number of letters in `{ccc}` corresponds to the number of columns. You can adjust this by adding more letters: `c` for center, `l` (lowercase L) for left, and `r` for right alignment within each column. Adding the pipe symbol (`|`) creates vertical lines between columns depending on its placement (e.g., `{|c|c|c|}` will draw lines on the outer edges and between every column).
- You must write the table caption inside `\caption{...}`. Remember to use `\protect` before any `\cite` commands if a citation is necessary within the caption.
- `\label{tab:yourlabel}` assigns a unique label to the table, allowing you to reference it in the text using `Table~\ref{tab:yourlabel}`.
- `\hline` draws a horizontal line. These are typically used to separate the header from the data and to close the table.
- `Column1 & Column2 & Column3 \\` places the data inside the cells. Each cell must be separated by an ampersand (`&`) and each row must end with a double backslash (`\\`) to proceed to the next line.
- `\end{tabular}` ends the table structure.
- `\end{table}` ends the allocated space for the table.

How to modify the timeline

To define a work package, use the following format:

```
\workpackage{Task Name}{Start Week}{End Week}
```

Example:

```
\workpackage{Literature Review}{1}{3}
```

This means the task will run from Week 1 through Week 3.

How to add references and citations

For in-text citations, you can use these commands:

- `\cite{example2020}` → Author (2020)
- `\citep{example2020}` → (Author, 2020)
- `\cite{example2020,example2021}` → Author (2020); Researcher (2021)
- `\citep[see][for details]{example2020}` → (see Author, 2020, for details)

You must place your reference information in the `references.bib` file using the following format:

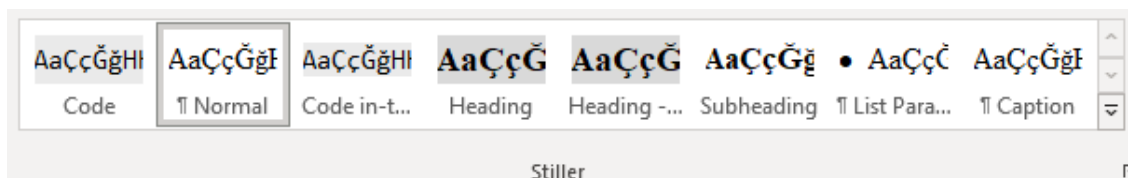
```
@article{example2020,
  author = {Author, A.},
  title = {Example Article},
  journal = {Journal},
  year = {2020},
  volume = {1},
  pages = {1-10}
}
```

- If you are using a local LaTeX compiler, you must compile `main.tex` with BibTeX to generate the references, then recompile with pdfLaTeX to produce the final PDF output (Overleaf performs this procedure automatically).
- Use the proper citation format via BibTeX; do not manually format your references.

Instructions for Microsoft Word

Styling

Standard formatting for headings and body text has been pre-set as Styles (“*Stiller*”) in the template. Simply select your text and apply the relevant style to ensure consistent formatting throughout your document.



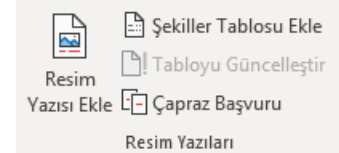
How to include a figure (recommended way)

- Place your image in the document.
- Click on  next to the image and select the “in line with text” (“*metinle aynı hizaya*”) option. You can then center the image using the same alignment tools as you would for text ().

- Right click the image and select “Insert Caption...” (“*Resim Yazısı Ekle...*”). The image captions should be **below** the image.
- Figure numbers in the captions will update automatically as you add more images with captions.

How to include a table caption

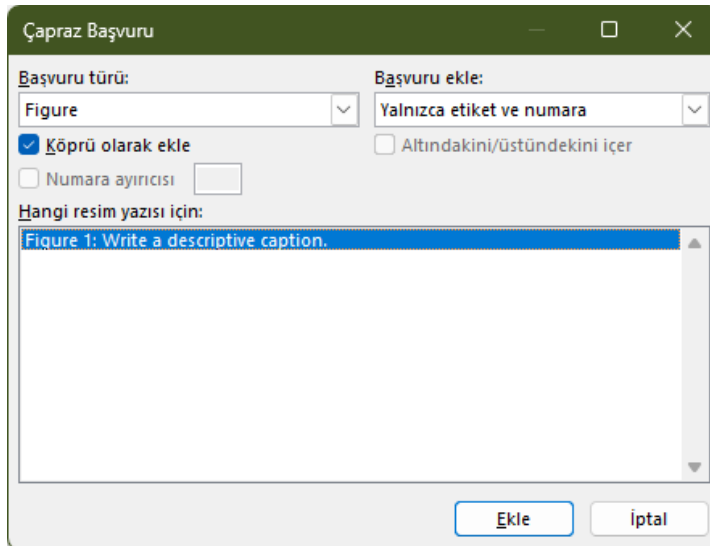
Similar to adding an image caption, right-click the table and select “Insert Caption...” (“*Resim Yazısı Ekle...*”). The table captions should be **above** the table.



How to add in-text cross-references to figures and tables (optional)

While you can type figure and table numbers manually, using the cross-reference tool ensures that numbers update when you insert or remove figures and/or tables later. These references are also clickable links that jump straight to the figure or table.

- First, add an image/table with a caption.
- When you want to reference it in the text, click on the “References” (“*Başvurular*”) tab at the top. Click on “Cross-reference” (“*Çapraz Başvuru*”).
- Choose the reference type (Figure or Table), insert reference for “Only label and number” (“*Yalnızca etiket ve numara*”). When added, the reference is added to the text.



- In-text references do not always update automatically. To manually update them, you can either:
 - Open the “Print” dialog (Ctrl + P), then close it; or
 - Select all text (Ctrl + A), right click any highlighted area, and select “Update Field” (“*Alanı Güncelleştir*”).